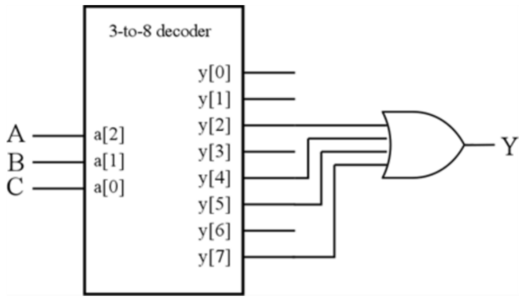
Tutorial 4: Combinational Logic Circuit – Building Blocks

Question 1

Realise the function $f_1(A,B,C) = \sum m(2, 4, 5, 7)$ with a 3-to-8 decoder and an OR gate.

Let $Y = f_1(A,B,C)$



Question 2

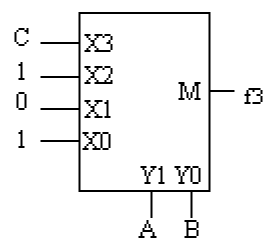
Realise the function $f_2(A,B,C) = (A + B')(B' + C)$.

$f_2(A,B,C) = (A + B')(B' + C) = \sum m(0, 1, 4, 5, 7)$

(i) 4-to-1 multiplexer

Let A and B = select signals.

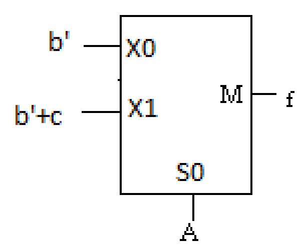
ABC	f_2	Mux input
000 001	1 1	$X_0 = 1$
010 011	0 0	$X_1 = 0$
100 101	1 1	$X_2 = 1$
110 111	0 1	$X_3 = C$



(ii) 2-to-1 multiplexer

Let A = select signal.

ABC	f_2	Mux input
000 001 010 011	1 1 0 0	$X_0 = b'$
100 101 110 111	1 1 0 1	$X_1 = (bc')'$ $= b'+c$



Question 3

An arithmetic logic unit (ALU) is a digital circuit used to perform arithmetic and logic operations. It represents the fundamental building block of the central processing unit (CPU) of a computer. A 32-bit ALU interface is shown in Figure Q3-1. The function table of the ALU is shown in Figure Q3-2.

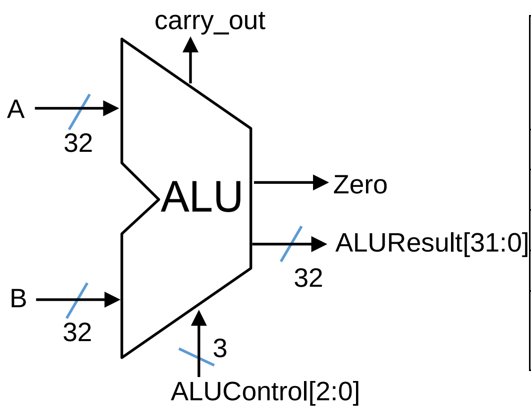
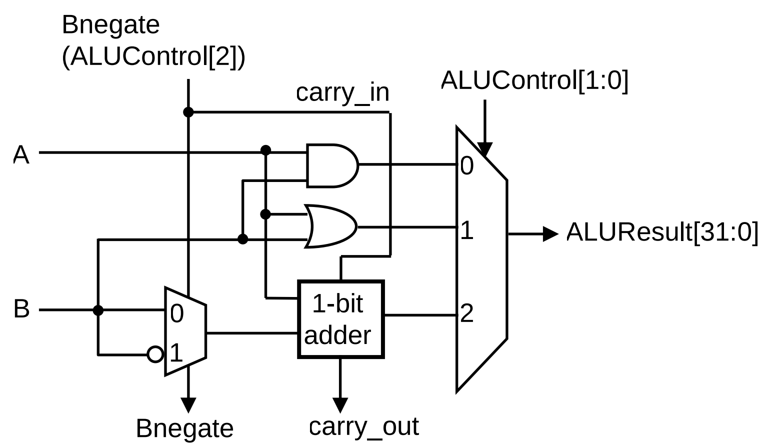


Figure Q3-1

ALU control input (ALUControl[2:0])	ALU action required	Function
000	and	and
001	or	or
010	add	add
110	sub	subtract
111	sub	set on less than

Figure Q3-2

(i) Draw the internal structure of the ALU cell which can support the functions and, or, add and subtract.



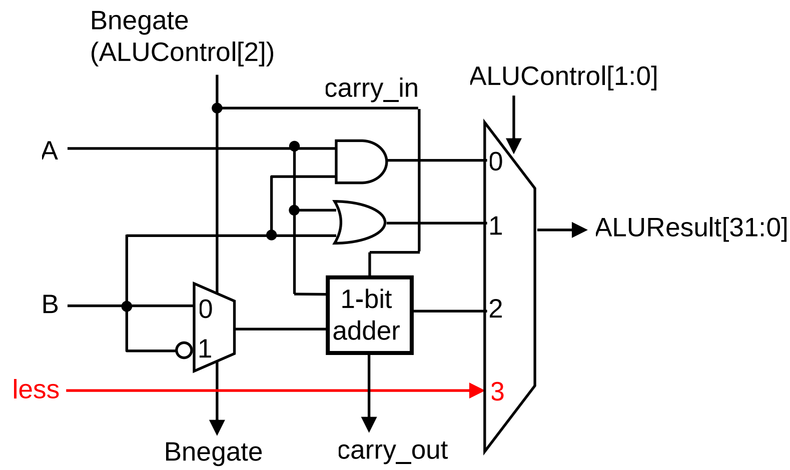
(ii) Modify the ALU cell you have developed earlier to support the “set on less than” function whereby if A < B, then ALUResult[31:0] = 1.

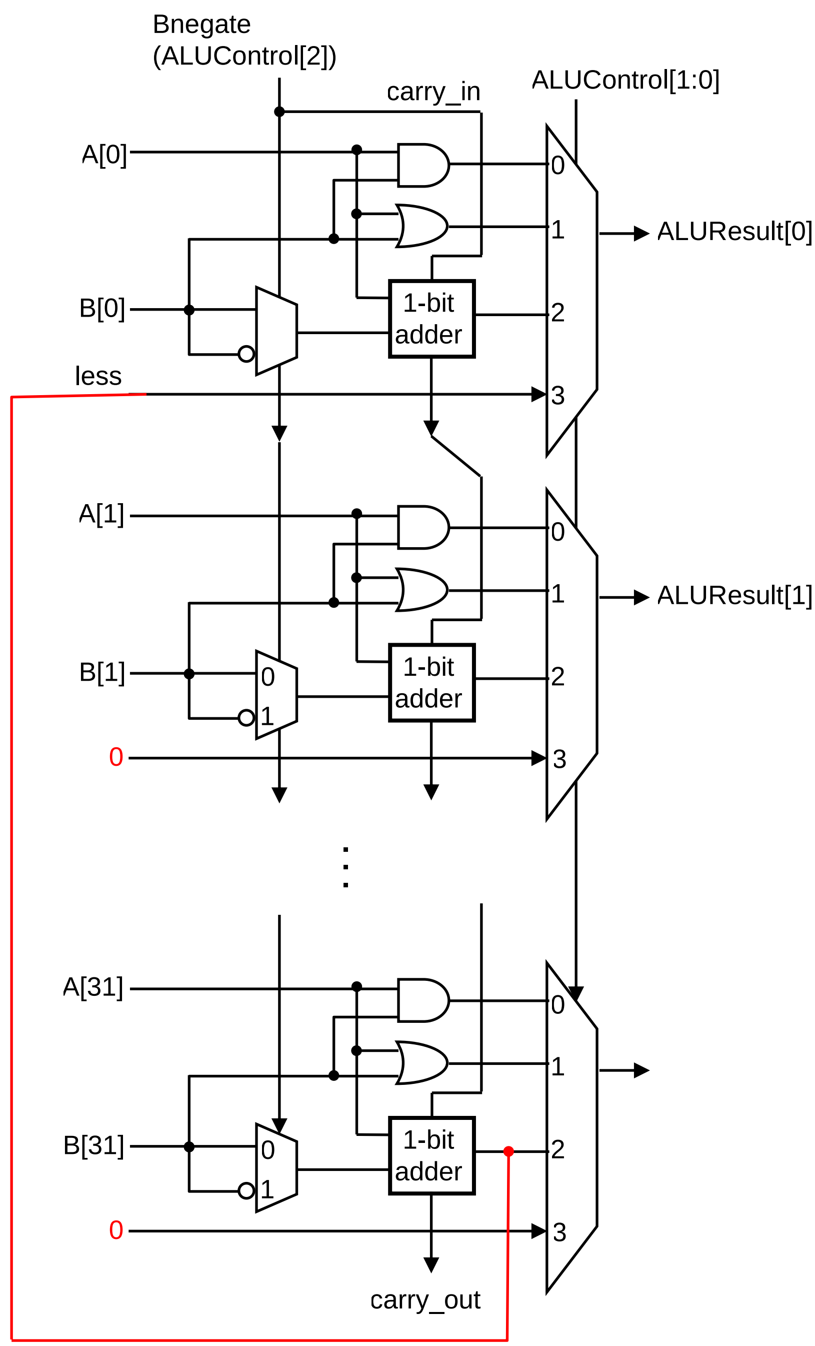
First perform a subtraction (A - B)

- If the resulting sign bit is negative, that means (A < B)
- If the resulting sign bit is positive, that means either (A > B) or (A == B)

We only need the sign bit to indicate “set on less than”

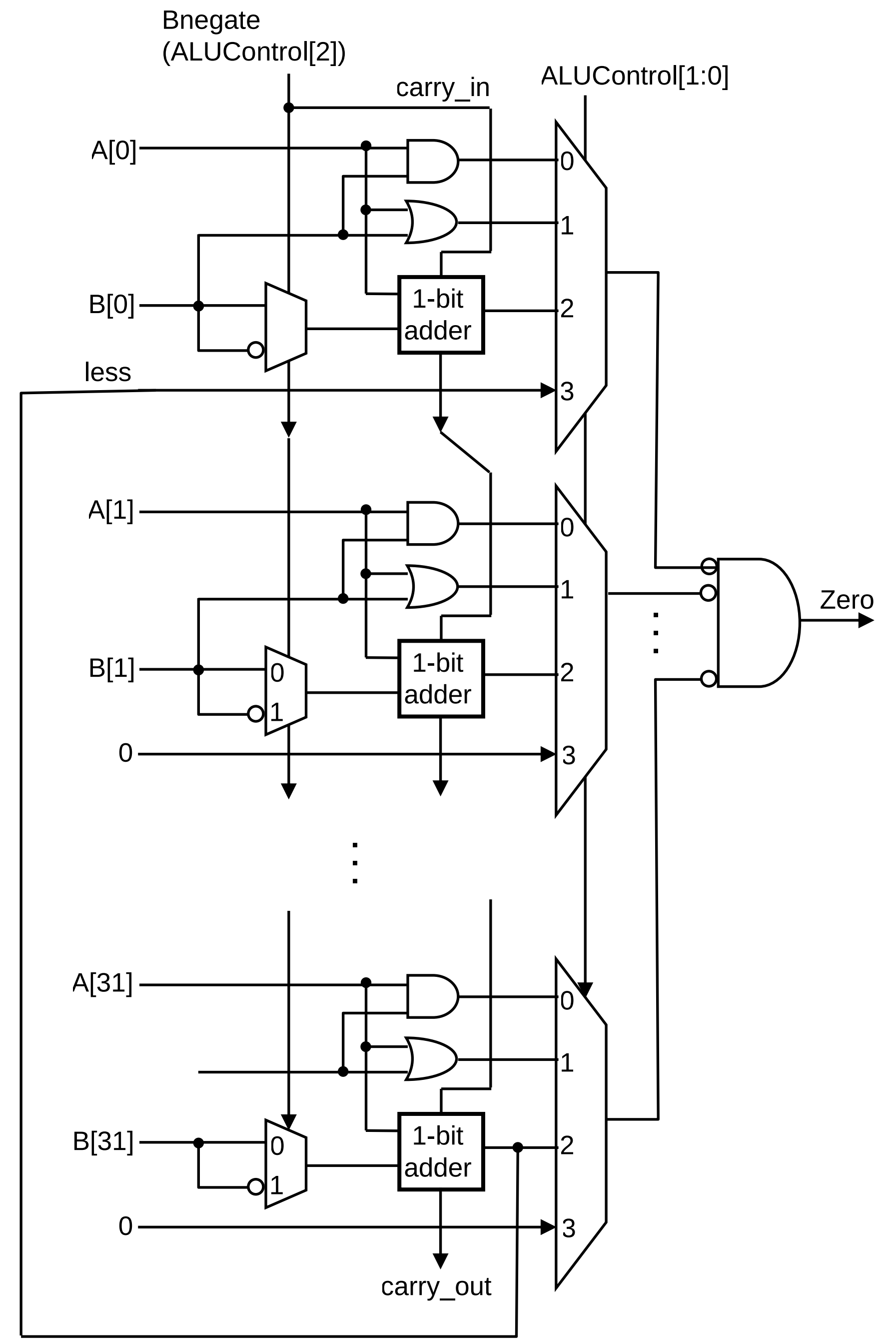
- Meaning connect the sign bit to the LSB of the result.


Bnegate
(ALUControl[2])
carry_in
ALUControl[1:0]
A
B
less
0
1
2
3
1-bit
adder
ALUResult[31:0]
Bnegate
carry_out


Bnegate
(ALUControl[2])
carry_in
ALUControl[1:0]
A[0]
B[0]
less
1-bit
adder
ALUResult[0]
A[1]
B[1]
0
1-bit
adder
ALUResult[1]
A[31]
B[31]
0
1-bit
adder
carry_out

(iii) Modify the ALU cell you have developed earlier to support “A == B”.

First perform subtraction
Insert additional logic to detect when all result bits are zero



Question 4

The following figure contains a positive-edge-triggered DFF and a negative-edge triggered DFF. Complete the timing diagram of the following figure by drawing the output waveforms for y[1] and y[2].